

Why Displacement Interpolation Gives Wasserstein Geodesics

Assume $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$ and that the optimal transport from μ to ν is induced by a map

$$T : \mathbb{R}^d \rightarrow \mathbb{R}^d, \quad T_{\#}\mu = \nu.$$

For the quadratic cost, define

$$T_t(x) = (1-t)x + tT(x), \quad t \in [0, 1],$$

and

$$\mu_t = (T_t)_{\#}\mu.$$

This curve moves each particle along the straight line from x to $T(x)$.

Why This Is a Geodesic

For $0 \leq s < t \leq 1$, the map sending $T_s(x)$ to $T_t(x)$ moves the same particle from its position at time s to its position at time t . The distance traveled by that particle is

$$|T_t(x) - T_s(x)| = |(t-s)(T(x) - x)| = |t-s| |T(x) - x|.$$

Therefore the quadratic transport cost between μ_s and μ_t is bounded by

$$W_2^2(\mu_s, \mu_t) \leq |t-s|^2 \int_{\mathbb{R}^d} |T(x) - x|^2 d\mu(x).$$

Since T is optimal from μ to ν ,

$$\int_{\mathbb{R}^d} |T(x) - x|^2 d\mu(x) = W_2^2(\mu, \nu).$$

Thus

$$W_2(\mu_s, \mu_t) \leq |t-s| W_2(\mu, \nu).$$

The reverse inequality follows from the triangle inequality:

$$W_2(\mu, \nu) \leq W_2(\mu, \mu_s) + W_2(\mu_s, \mu_t) + W_2(\mu_t, \nu).$$

Using the same upper-bound argument on the intervals $[0, s]$ and $[t, 1]$ gives

$$W_2(\mu, \mu_s) \leq s W_2(\mu, \nu), \quad W_2(\mu_t, \nu) \leq (1-t) W_2(\mu, \nu).$$

Combining these inequalities forces

$$W_2(\mu_s, \mu_t) \geq |t-s| W_2(\mu, \nu).$$

Together,

$$W_2(\mu_s, \mu_t) = |t-s| W_2(\mu, \nu).$$

This is exactly the constant-speed geodesic condition.

Interpretation

Euclidean geodesics move points along straight lines. Wasserstein geodesics move mass along optimal straight-line particle trajectories. This is why the curve $\mu_t = (T_t)_{\#}\mu$ is called displacement interpolation.